ECON 601 Final Research Proposal

To: Prof. Steve Berry & Prof. Katja Seim
From: Marek Chadim
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Title: The Dynamics of Market Power in Public Procurement: A Structural Estimation of Entry Barriers and Markup Premiums

Note: This proposal extends my reduced-form MSc thesis (Chadim 2025, SSE) by moving from causal treatment-effect estimation to a structural dynamic discrete-choice framework, layered over a first-price sealed-bid procurement auction in the spirit of Kroft, Luo, Mogstad, and Setzler (2025).

1 Research Question

Public procurement constitutes approximately 12% of GDP across OECD countries, representing $1.3 trillion annually. Despite its macroeconomic significance, the interaction between public procurement design and firm-level pricing power remains understudied. In my previous work (Chadim, 2025), I utilized a novel panel dataset of Czech construction firms and applied modern causal inference estimators (Synthetic Difference-in-Differences and Matrix Completion). I established a robust stylized fact: firms experience a 15% causal increase in markups upon entering the public procurement market. Furthermore, this premium declined dramatically from 30% in 2006 to 10% in 2021, coinciding with the 2012 Czech reform that banned single-bidding procedures.

While these reduced-form estimates quantify *that* a markup premium exists and is highly sensitive to institutional design, they cannot identify the underlying structural mechanisms sustaining this market power. If government contracts are highly lucrative, standard economic theory dictates that entry should drive the premium to zero. The persistence of this premium suggests significant frictions on *two* margins simultaneously: (i) the *entry* margin, where prospective bidders face sunk costs of qualifying for tender processes, and (ii) the *auction* margin, where conditional on entry, asymmetric information and strategic bidding generate within-tender markups.

Research Question: What are the structural sunk costs of entering the public procurement market, and to what extent do these barriers to entry sustain the observed markup premium by shielding incumbents from competition? Conditional on entry, how much of the observed premium is generated by asymmetric bidding within auctions?

Importance: Answering this question is critical for procurement policy. If the 15% markup premium is sustained by massive, unrecoverable sunk costs of navigating bureaucratic tender processes, then simply banning single-bidding (as done in 2012) is insufficient. To fully optimize taxpayer funds, policymakers need to know whether subsidizing entry costs (e.g., simplifying tender applications for SMEs) would increase competition enough to drive down procurement markups, and what the net welfare effect of such a policy would be. A second motivation is method-

ological: in a contemporaneous structural study of the U.S. construction industry, Kroft, Luo, Mogstad, and Setzler (2025, hereafter KLMS) prove (their Propositions 1–2) that the wage markdown and price markup are *multiplicatively*, not additively, related—the optimal price satisfies $P_{jt} = (1-\epsilon)^{-1}(1+\theta) \cdot W_{jt}/MPL_{jt}$, so the "double markup" relative to productivity-adjusted wage is the product of the product-elasticity channel $(1-\epsilon)^{-1}$ and the labor-elasticity channel $(1+\theta)$. Empirically, KLMS estimate the product-only markup at 16% (almost identical to my Czech reduced-form 15%) and the labor-only markdown at 22%, but the structurally-corrected double markup is 44–49%. This raises the possibility that my reduced-form result captures only the product-elasticity channel and substantially understates the true degree of market power once the multiplicative interaction with labor-market power is accounted for.

2 Data

I will leverage and expand upon the dataset constructed for my previous research. The data consists of an unbalanced panel of 1,297 Czech construction firms tracked over 16 years (2006–2021), totaling 7,261 firm-year observations.

The dataset is constructed by linking two primary sources:

1. **Firm Financial Statements (MagnusWeb):** Provides granular, firm-level balance sheet data including revenues, cost of goods sold (COGS), labor, and capital assets. I use this data to estimate firm-specific markups and unobserved total factor productivity (ω_{it}) using the De Loecker and Warzynski (2012) production function approach.

2. **Public Procurement Records (Datlab s.r.o.):** Provides contract-level details (values, dates, awarding entities) matching firm IDs. The thesis aggregated this source to firm-year participation, but the underlying data is at the tender-bid-firm level—containing the bidder identities and bid amounts required for structural auction estimation in the spirit of Guerre, Perrigne, and Vuong (2000) and KLMS (2025).

Key Data Structure for Structural Estimation:
The critical variation for this proposed structural model lies in the transition dynamics. In the data, 44% of observations involve government contracting. Crucially, I observe firms that continuously operate in the private sector, firms that transition into the public sector (paying the sunk cost of entry), firms that exit the public sector (saving the fixed cost of participation), and continuous contractors. This rich matrix of entry and exit decisions, conditional on the firm's evolving productivity and capital stock, provides the necessary variation to identify structural cost parameters. At the auction level, the contract-level Datlab records additionally provide the bid distributions required to invert the bid function and recover latent cost distributions.

Preliminary descriptive evidence from Datlab. The contract-level Datlab data already constructed for the thesis comprises 233,870 Czech construction tenders (2006–2021) with bidder

counts. Figure 1 plots three competitive-intensity outcomes by year, with 95% bootstrap confidence bands and dashed reference lines at the 2012 single-bidding ban and the 2017 Register of Contracts:

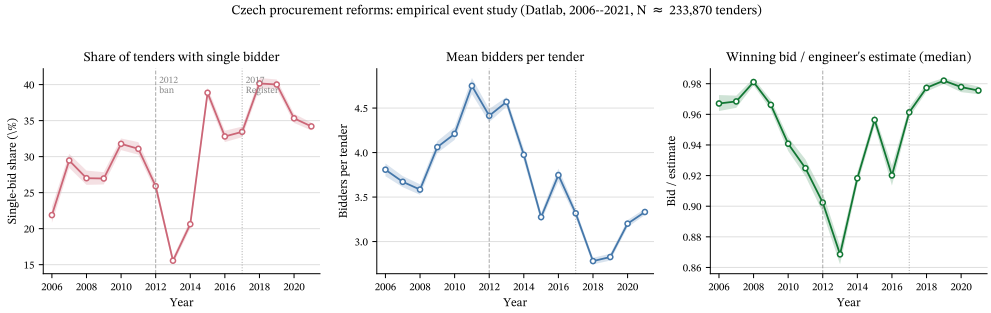


Figure 1: Czech procurement reforms produce a transient drop in single-bidding (2013 trough at 16%) but a persistent decline in mean bidders per tender (4.1 → 2.8 by 2018), suggesting reduced-form reform effects are complicated and motivating structural decomposition. Source: Datlab, $N = 233{,}870$ tenders.

The 2012 ban initially halved the single-bid share (29% → 16% by 2013) but the system reverted to 34–40% by 2018–2021; the mean bidders per tender fell from 4.06 (pre-2012) to 3.51 (post-2012), a 14% decline; and the median winning bid moved from 96% to 95% of the engineer's estimate. The reduced-form reform impact is therefore real but transient and partial—exactly the kind of phenomenon that requires structural decomposition (§4 below) to interpret as a movement in entry-cost frictions, asymmetric-information rents, or auction-design substitution. Following Haile's (2024) descriptive–structural taxonomy, Figure 1 and the thesis (Chadim 2025) are *descriptive*—characterizations of relationships among observables; the model in §3 and the recovered cost distributions in §4 are *structural*—features of an assumed data-generating process whose recovery is what makes counterfactual policy simulation possible.

3 Model

To explain firm participation in public procurement and the markups conditional on participation, I propose a two-layer structural model. The outer layer is a dynamic discrete choice model of entry and exit (Rust, 1987; Ericson and Pakes, 1995). The inner layer is a first-price sealed-bid procurement auction with type-asymmetric bidders, in the spirit of KLMS (2025) Section I and the broader Guerre–Perrigne–Vuong (2000) literature. The two-layer separation builds on Jofre-Bonet and Pesendorfer (2003), the canonical structural treatment of forward-looking bidders in U.S. highway construction, who model auctions as a unified dynamic game with capacity-constrained bidders; relative to JBP, I move the dynamics from *within* the auction (capacity) to the *participation* margin (entry/exit), the variation that Czech reform-era data (2012 single-bidding ban, 2017 Register of Contracts) identifies most cleanly. Crucially, the expected per-period procurement profit that enters the entry game's Bellman equation is no longer reduced-form; it is structurally derived from the equilibrium of the auction game.

3.1 Outer Game: Dynamic Entry/Exit

In each period t, firm i observes its state vector $s_{it} = (\omega_{it}, k_{it}, W_{i,t-1})$.

- ω_{it} is the firm's productivity (recovered from the DLW production function estimation).
- k_{it} is the firm's capital stock.
- $W_{i,t-1} \in \{0, 1\}$ is the firm's procurement participation status in the prior year (Incumbent vs. Potential Entrant).

The firm chooses an action $a_{it} \in \{0, 1\}$, representing whether to participate in public procurement in year t (i.e., bid for at least one contract).

Per-Period Payoff:

The firm earns baseline profits from the private market $\pi^{priv}(s_{it})$. If $a_{it} = 1$, the firm additionally earns expected procurement profits $\pi^{pub}(s_{it})$ derived from the auction game described in §3.2. However, participation is costly.

- If $a_{it} = 1$ and $W_{i,t-1} = 0$, the firm pays a **Sunk Entry Cost (S)** to establish bidding infrastructure, learn the legal framework, and build political/institutional capital.
- If $a_{it} = 1$ and $W_{i,t-1} = 1$, the firm pays a **Fixed Participation Cost (F)** to maintain its bidding department.

The total per-period utility is given by:

$$U(a_{it}, s_{it}) = \pi^{priv}(s_{it}) + a_{it} \cdot \Big[\pi^{pub}(s_{it}) - S \cdot (1 - W_{i,t-1}) - F \cdot W_{i,t-1}\Big] + \epsilon_{it}(a_{it})$$

Where ϵ_{it} is an i.i.d. Type-1 Extreme Value shock representing idiosyncratic, unobserved cost/revenue shocks to participation.

Value Function:

Firms are forward-looking and maximize the expected present discounted value of lifetime profits. The Bellman equation is:

$$V(s_{it}, \epsilon_{it}) = \max_{a_{it} \in \{0,1\}} \{U(a_{it}, s_{it}) + \beta \, E[V(s_{i,t+1}, \epsilon_{i,t+1}) \mid s_{it}, a_{it}]\}$$

Where β is the discount factor, and the expectation is taken over the transition probabilities of productivity, capital, and the realized auction sequence in $t + 1$. Relative to KLMS (2025), who study a single-period bidding decision, my outer game makes participation a forward-looking decision driven by the staggered Czech procurement reforms (notably the 2012 single-bidding ban and the 2017 transparency requirements), giving identification leverage that a static model cannot exploit.

3.2 Inner Game: First-Price Sealed-Bid Procurement Auction

Conditional on $a_{it} = 1$, the firm bids for individual contracts in a sequence of first-price sealed-bid auctions, following KLMS (2025) Section I.B and the canonical structural-auction literature. KLMS show that the production side admits a closed-form composite production function $Q_{jt} = \min\{\Phi_{jt} L_{jt}^{\rho}, \beta_M M_{jt}\} \exp(e_{jt})$ with composite returns to labor $\rho \equiv (1+\theta)\beta_K + \beta_L$ — encoding the labor-market power θ inside the production technology so that ρ becomes the natural object the auction stage identifies. I adopt this composite-returns formulation and extend it by endogenizing the binary participation indicator a_{it}. For a tender t with n_t participants, each bidder i draws a private cost c_{it} from a type-specific distribution $F_C(\cdot \mid \tau_i, X_t)$, where:

- $\tau_i \in \{\text{incumbent}, \text{entrant}\}$ indexes the firm's procurement history—the asymmetry the proposal seeks to identify.
- X_t is a vector of tender characteristics (engineer's estimate, NACE 41/42/43 sub-industry, awarding entity, urban/rural location).

The firm submits an equilibrium bid $\beta_\tau^*(c_{it} \mid n_t, X_t)$ characterized by the asymmetric first-price first-order condition (Lebrun 1999; Maskin and Riley 2000; Bajari 2001):

$$\beta_\tau^{*\prime}(c) \;=\; (n_t - 1) \cdot \frac{f_{-\tau}\big(\phi_{-\tau}(\beta_\tau^*(c))\big)}{1 - F_{-\tau}\big(\phi_{-\tau}(\beta_\tau^*(c))\big)} \cdot \big(\beta_\tau^*(c) - c\big), \quad \beta_\tau^*(\overline{c}) = \overline{c},$$

where $\phi_\tau \equiv (\beta_\tau^*)^{-1}$ is the inverse-bid function. Type asymmetry $F_C^{\text{inc}} \neq F_C^{\text{ent}}$ generates the central testable implication: the weaker type bids more aggressively (closer to cost), the stronger type less aggressively. If it wins, the firm executes the contract at marginal cost c_{it}, earning the realized markup $\beta_\tau^*(c_{it}) - c_{it}$. The expected procurement profit that enters the outer game's Bellman equation is therefore:

$$\pi^{pub}(s_{it}) = N_t \cdot E\left[(\beta^*(c \mid \tau_i, n, X) - c) \cdot \Pr(\text{win} \mid c, \tau_i, n, X)\right],$$

where N_t is the firm's annual auction volume and the expectation integrates over (c, n, X) given the firm's state. This closes the model: entry decisions are forward-looking over an explicit auction equilibrium with type-asymmetric bidders, and the 15% markup premium is decomposed into (a) an *entry-margin* component, sustained by S and F, and (b) an *auction-margin* component, sustained by the gap between incumbent and entrant cost distributions.

3.3 Equilibrium Concept and Identifying Assumptions

The model's equilibrium has two interlocking layers. In the outer game, I impose *Markov-perfect equilibrium* (Maskin and Tirole 1988; Pakes and McGuire 1994): each firm's policy is a best response to rivals' policies given only the payoff-relevant state s_{it}, so multiplicity from history-dependent strategies is ruled out. In the inner auction game, I impose *Bayesian Nash equilibrium* under

independent private values (IPV): conditional on (τ_i, X_t), each firm's cost draw is independent of rivals' draws, and bid functions are strictly increasing in cost. The IPV restriction is the standard identification-enabling assumption in the GPV (2000) literature; it would be violated in pure common-values settings (which are unlikely in standardized construction tenders, since unit-cost components such as cement, rebar, and labor are largely public information after engineer estimates are published).

The two layers are linked by an envelope condition: the value-function derivative with respect to participation status equals the difference between (i) the auction-derived expected profit minus participation cost, and (ii) the option value of remaining a non-participant. This connection means parameters (S, F) from the outer game and $F_C(\cdot \mid \tau)$ from the inner game are *separately* identified rather than jointly: I can estimate them sequentially in three steps without solving a joint fixed point. Three additional assumptions discipline the estimation: (a) firms do not collude across auctions (no bid-rotation cartels, an assumption Baránek and Titl 2024 partially test by showing connection-driven price premia are concentrated within rather than across firms); (b) the type indicator $\tau_i \in \{\text{incumbent}, \text{entrant}\}$ is observable to bidders within a tender, which is institutionally plausible in Czech procurement where bidder histories are public; (c) the discount factor $\beta = 0.95$ is calibrated rather than estimated, following the standard practice in dynamic-game estimation (Bajari, Benkard, and Levin 2007).

Defending IPV. IPV is testable: following Athey and Haile (2002, 2007) and Haile, Hong, and Shum (2003), I will check (i) bid-distribution monotonicity in n_t, (ii) residualised within-tender bid independence, and (iii) rationalisability of empirical bids by some primitive cost distribution. If the tests reject pure IPV, Krasnokutskaya (2011) provides a robustness path through multiplicative auction-level unobserved heterogeneity $c_{it} = u_t \cdot v_{it}$ recovered by nonparametric deconvolution.

4 Methods and Identification

Estimating this two-layer model via full-solution Maximum Likelihood is computationally infeasible given the continuous state variables (ω, k) and the auction equilibrium fixed point. I therefore use a sequential three-step estimator that combines structural auction estimation (Step 0) with the two-step Bajari, Benkard, and Levin (2007) / Aguirregabiria and Mira (2007) approach to dynamic games (Steps 1–2).

Step 0: Auction-Stage Estimation.

I use the contract-level Datlab data—which the thesis collected but only used at the firm-year aggregate—to recover the latent type-specific cost distributions $F_C(\cdot \mid \tau_i, X_t)$ via two complementary identification strategies. *(a) Guerre, Perrigne, and Vuong (2000) nonparametric inverse-bid:* for each tender I observe the winning bid (and the runner-up bids where Datlab records them), the number of bidders n_t, and the tender characteristics X_t. The GPV inverse-bid mapping

$$\hat{c}_i = \beta_i - \frac{\hat{F}_\beta(\beta_i)}{(n-1) \cdot \hat{f}_\beta(\beta_i)},$$

applied separately for incumbent and entrant types, recovers $\hat{F}_C(\cdot \mid \tau, X)$ without parametric assumptions. As a proof-of-concept, Figure 2 reports a pilot implementation on the Datlab 2010–2018 subsample: incumbent and entrant bid distributions are nearly identical in raw bids (left panel), but the structurally-recovered cost distributions reveal an asymmetry of ∼4 percentage points (incumbents median cost 0.747 of engineer estimate vs entrants 0.791; right panel). Median markups follow the asymmetric-IPV prediction: stronger types extract more rent (incumbents 7.8% vs entrants 4.8%).

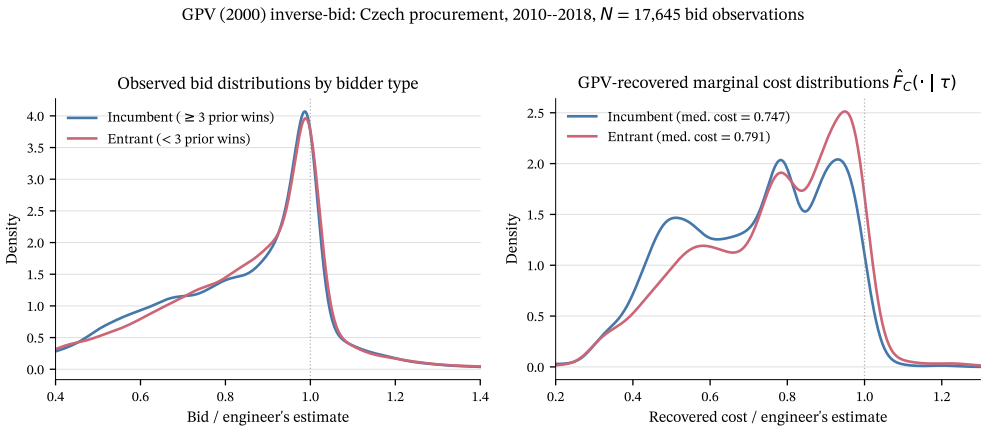


Figure 2: Pilot GPV (2000) recovery on Datlab 2010–2018 (N = 17,645 bid observations after positivity filter). Left: observed bid distributions by bidder type (incumbent = ≥3 prior wins). Right: recovered cost distributions $\hat{F}_C^I, \hat{F}_C^E$. Asymmetry (∼4pp cost gap, ∼3pp markup gap) is consistent with KLMS-style strong/weak type asymmetry.

(b) KLMS (2025) bid-monotonicity control function (Proposition 5): the equilibrium bid function is monotone in productivity given amenities, $s_u(\phi_{jt}) = \sigma_u(\phi_{jt}) \cdot \delta_u(\phi_{jt})$ with $\delta_u \geq 1$, so the inverse $\phi_{jt} = s_u^{-1}(Z_{jt})$ recovers unobserved TFP from the observed bid Z_{jt}. This is the methodological pearl of KLMS: bids serve as a control variable for the unobserved-productivity confounder in production-function estimation, side-stepping the standard simultaneity bias. I reuse this trick for the Czech sample. *(c) Winners-vs-losers matching*: for tenders with multiple close bids, I compare firm-level outcomes (revenue, employment, contract markup) of the winner against the runner-up. The identifying assumption (KLMS Proposition 3) is that the bidder-firm idiosyncratic labor-supply shock $\Delta\nu_{jt}$ is unpredictable at the time of bid submission—justified institutionally by the time delay between bid solicitation and production commencement—so winner status D_{jt} is independent of $\Delta\nu_{jt}$ in the differenced wage and employment equations. *(d) Bajari and Hortaçsu (2005):* a parametric structural-auction specification provides robustness to the GPV nonparametrics and a benchmark of structural fit. *(e) Athey, Levin, and Seira (2011):* their comparison framework for sealed-bid versus ascending-bid auctions enables auction-format counterfactuals. *(f) Krasnokutskaya (2011) deconvolution:* to address tender-level unobserved heterogeneity (project complexity, awarding-entity scale) that would otherwise contaminate the GPV inversion, I write $c_{i,t} = u_t \cdot v_{i,t}^{\tau_i}$ with u_t a multiplicative auction-level component, recoverable by characteristic-function

deconvolution under multiplicative independence; the published engineer's estimate—available for every Datlab tender—additionally serves as a control-function proxy for u_t in the spirit of Athey and Haile (2007 *Handbook*, §3.4.2). Bidder-type-specific cost distributions F_v^τ are then identified from demixed bid residuals.

Step 1: Estimate Policy Functions and State Transitions.
Using the structural $\hat{\pi}^{pub}(s_{it})$ from Step 0, I estimate the state transition probabilities $P(s_{i,t+1} \mid s_{it}, a_{it})$ directly from the data (e.g., modeling productivity ω as an AR(1) process). Concurrently, I estimate the Conditional Choice Probabilities (CCPs)—the empirical probability that a firm chooses $a_{it} = 1$ given its state s_{it}—using a flexible logit model or generalized random forest.

Step 2: Forward Simulation and Parameter Estimation.
Using the estimated CCPs, state transitions, and auction-derived expected profits, I simulate the expected present discounted value of lifetime profits for a given set of structural parameters (S, F). I then search for the values of S and F that make the observed firm decisions optimal, minimizing profitable deviations.

Identification. For each of the five structural primitives $(S, F, F_C^{\text{inc}}, F_C^{\text{ent}}, \rho_\omega)$ plus the calibrated discount factor β, the identifying moment and the Czech reform-IV exclusion restriction are summarized in the *Identification Summary* at the end of the main proposal. The reform IVs are *quasi-experimental within structural* in the sense Haile (2024) clarifies: they provide credible cross-time variation that identifies structural features of the model, not an alternative to structural estimation. Each row of the summary links a primitive to a moment in the data and a maintained exogeneity condition the reform supplies.

What would break identification. Three threats are foreseeable. First, if reform timing correlates with unobserved cost shifters (e.g., commodity-price shocks coincident with 2012), the reform-IV exclusion fails; I will instrument with the post-2012 dummy interacted with EU-level commodity prices to absorb this channel. Second, if the type indicator τ_i is misclassified (an "entrant" who has informally bid through a parent firm), F_C^{inc} and F_C^{ent} are partially set-identified rather than point-identified; bounds in the spirit of Haile-Tamer (2003) will be reported. Third, if firms collude across auctions, the bid FOC fails; I will run the Bajari and Ye (2003) competition-vs-collusion test and Decarolis-Giorgiantonio (2022) red-flag indicators on the Czech sample as a precondition.

Counterfactual Simulations.
Once the structural primitives (S, F, F_C) are identified, I run three counterfactual policy experiments. (i) *Entry-cost subsidy.* The government subsidizes the sunk entry cost S by 50% for all firms; I recompute optimal entry policies and the stationary mass of procurement participants, and map the increased competitive density back into aggregate auction-stage markups. (ii) *Auction-format reform.* Switching from first-price sealed-bid to ascending auctions following Athey, Levin, and Seira (2011), holding the firm population fixed; this isolates the within-auction component of the procurement premium. (iii) *Reserve price plus SME set-aside.* Implementing the EU Procurement Directive 2014/24/EU recommendations as a counterfactual auction design; this combines the

entry margin and auction margin in a single policy and is closest to the actual reform space available to Czech and EU regulators. Together, these three experiments decompose the 15% Czech procurement-markup premium into an entry-friction component, an auction-friction component, and a residual interpreted as quality differentiation, providing the policy answer the reduced-form thesis cannot deliver.

5 Expected Contribution

Contribution relative to KLMS (2025). The closest published paper, KLMS (2025), studies imperfect competition in U.S. construction with a two-stage auction-then-production model but treats firm participation D_{jt} as the exogenous auction outcome, ignores forward-looking dynamics, treats firm output as a scalar (inheriting the Pakes 2021 multiproduct critique), and does not exploit any policy-reform variation. My proposal extends KLMS in four directions: (i) the binary participation indicator is endogenized via a Rust/Ericson–Pakes dynamic-discrete-choice outer game with sunk cost S and fixed cost F; (ii) Czech procurement reforms (2012, 2017) supply quasi-experimental exclusion restrictions absent in the KLMS U.S. observational design (§4 identification table); (iii) the contract-level Datlab data permits subindustry decomposition of $(\epsilon, \beta_L, \rho)$ across NACE 41/42/43, addressing the Pakes (2021) multiproduct critique that KLMS leaves open—a preliminary thesis-pipeline check shows the procurement-markup premium is qualitatively similar across construction sub-industries ($\hat{\beta}_{pp} \in [0.110, 0.163]$ across NACE 41/42/43 with cluster-firm SEs < 0.01, $N = 7{,}115$), necessary but not sufficient evidence that the multiproduct critique is empirically muted; (iv) cross-country external validity—the Czech 30%/45% magnitudes (under structural recovery) provide the first non-U.S. test of KLMS's headline finding outside the U.S. construction-industry setting.

Contribution relative to Baránek and Titl (2024). Baránek and Titl establish a 6% favoritism premium via close-election RDD; my proposal complements their reduced-form result by structurally decomposing the 15% premium into entry-friction and auction-friction components, two channels their RD cannot separately identify (developed in the appendix referee report).

Identification Summary

Each of the five structural primitives is paired with an identifying moment in the data and a Czech reform-IV exclusion restriction supplying cross-time variation. The discount factor β is calibrated rather than estimated, with sensitivity reported.

Primitive	*Identifying moment*	*Reform-IV exclusion restriction*
S (sunk entry cost)	Cross-firm difference in entry rates at given (ω, k) between incumbents ($W_{i,t-1} = 1$) and non-incumbents ($W_{i,t-1} = 0$); cutoff productivity ω_t^* at which a non-incumbent is just indifferent between entering and not.	Act 55/2012 (single-bidding ban) shifts the cutoff ω_t^* discontinuously; the post-2012 jump in non-incumbent entry pins S. Exclusion: $E[\eta \mid \mathbb{1}\{t \geq 2012\}] = 0$ within the productivity-controlled subpopulation.
F (fixed continuation cost)	Exit rate of incumbents conditional on $(\omega, k, W_{i,t-1} = 1)$; the survival hazard at low productivity.	Act 134/2016 (MEAT criteria) raises documentation cost and identifies F via post-2016 exit rates among incumbents with declining ω.
$F_C^{\text{inc}}, F_C^{\text{ent}}$ (type-specific cost distributions)	Bid distributions $\{b_{i,t}\}$ separately for incumbent and entrant types within each X_t cell, inverted via GPV.	2017 Register of Contracts narrows incumbent informational advantage; pre/post-2017 change in bid-distribution dispersion identifies the information-asymmetry component of $F_C^{\text{inc}} - F_C^{\text{ent}}$.
ρ_ω (productivity persistence)	Within-firm AR(1) lag-coefficient on ω_{it} residualised against capital and tender-flow shocks.	Standard panel-AR(1) on the firm-year MagnusWeb sample; no reform IV required.
β (discount factor)	Calibrated to 0.95 (annual).	Robustness: $\beta \in \{0.90, 0.95, 0.99\}$.

Appendix: Referee Report

The Cost of Favoritism in Public Procurement

Baránek and Titl (2024)

Referee: Marek Chadim

Summary

This paper estimates the causal effect of political connections on public-procurement outcomes in the Czech Republic via close-election RDD on 1,500-plus Czech municipal elections (2010–2018), linked to the universe of municipal procurement contracts (>250,000 contracts) and the corporate political-donations register. The headline finding (Table 3, col. 3): contracts awarded to firms connected to the narrowly-winning party are priced 6.05% higher (s.e. 1.85%), with no offsetting quality improvement on observable margins (delays, change orders, completion rates). A complementary heterogeneity result (Table 6) shows the favoritism premium is roughly halved under EU co-financing oversight, establishing that institutional design moderates the rent-extraction channel.

Contribution relative to the literature: The paper sits between the cross-country corruption-index tradition and the structural-IO procurement program of Krasnokutskaya (2011) and Krasnokutskaya and Seim (2011). It operationalises "favoritism" as a quantifiable price wedge—a sharper construct than corruption indices, but mechanism-silent relative to a structural auction model. The closest companion identification strategy is the procurement red-flag detection program of Decarolis and Giorgiantonio (2022). The paper's distinctive contribution is the conjunction of (i) a credible RD on connection status, (ii) the comprehensive donations–elections–contracts data linkage, and (iii) the EU co-financing heterogeneity result, isolating the institutional moderator of the rent-extraction channel.

Two caveats on framing. First, the 6.05% premium is mechanism-silent: bid-rigging, information asymmetry, and entry deterrence are observationally equivalent at the reduced-form level but have different welfare and policy-design implications. Second, the estimand is a Local Average Treatment Effect at the close-election cutoff. The Stigler–Peltzman tradition suggests deeply entrenched coalitions extract larger rents than narrowly-elected ones, so the LATE may understate the ATT for the median connected firm.

The identification design is exemplary, and the substantive contribution is sharp. However, I am frustrated by the gap between the reduced-form result and the underlying mechanisms, and concerned about the LATE estimand and the omission of post-2017 transparency reforms.

Essential Points

1. **Mechanism Silence: Bid-Rigging vs. Information vs. Entry Deterrence.** The 6.05% premium is consistent with three distinct equilibrium mechanisms—bid-rigging cartels, infor-

mation asymmetry, and entry deterrence by connected incumbents—each with different policy implications. The reduced-form RD cannot distinguish them.

Concrete step forward: A structural auction with asymmetric bidders identifies the channels separately. Following Krasnokutskaya (2011), write

$$\log b_{i,t} \;=\; \log u_t \;+\; \log v_{i,t}^{\tau_i}, \qquad \tau_i \in \{\text{conn}, \text{unconn}\},$$

with u_t an auction-level unobserved component and $v_{i,t}^{\tau}$ the type-specific bidder-private cost. Multiplicative-independence deconvolution identifies F_u and $\{F_v^{\text{conn}}, F_v^{\text{unconn}}\}$ separately; Guerre, Perrigne, and Vuong (2000) inverse-bid then recovers marginal cost type-by-type, decomposing the 6.05% premium into a *latent-cost-asymmetry* component versus an *information-rent* component. Krasnokutskaya and Seim (2011) provide a published template for this entry-plus-asymmetric-auction architecture.

2. **LATE versus ATT and the Stigler–Peltzman Margin.** The close-election RDD identifies a LATE at the cutoff: the marginal effect when the connected party narrowly wins. Stigler–Peltzman predicts deeply entrenched coalitions extract larger rents than those teetering at the close-election margin, so the 6.05% LATE *understates* the ATT for the median connected firm.

   Concrete step forward: Estimate separately on (a) close-margin vs. landslide elections, and (b) by tenure of the connected coalition. If the close-margin estimate is smaller, report bounds on the ATT under monotonic extrapolation in margin and re-frame the headline as a lower bound.

3. **Static Framework Misses Dynamic Channels.** The static framework leaves two dynamic channels unmodelled: connected winners accumulate productivity advantages over time (Ferraz, Finan, and Szerman 2015), and entry deterrence shrinks the competitive density of subsequent auctions. Both make 6% a lower bound on the full welfare cost of favoritism.

   Concrete step forward: A two-step BBL (Bajari, Benkard, and Levin 2007) / Aguirregabiria and Mira (2007) estimator on the bidder-firm panel can simulate the counterfactual "connections blindly masked" and quantify entry-deterrence efficiency loss separately from the static price wedge (Roberts 2010; Athey, Levin, and Seira 2011).

Suggestions (Free Disposal)

1. **Standard RDD diagnostics.** Present McCrary (2008) density tests at the close-election cutoff to rule out manipulation of the running variable, and follow Gelman and Imbens (2019) in restricting polynomial order in the local-linear specification. The current robustness battery does not display the smoothed density plot, leaving the no-manipulation assumption untested rather than tested-and-validated.

2. **Functional-form sensitivity of parallel trends.** The event-study appendix specifications implicitly invoke a parallel-trends-like assumption on connected-vs.-unconnected firm price levels. Roth and Sant'Anna (2023) demonstrate that parallel trends can hold under one functional form (levels) while failing under another (logs). Robustness of the 6.05% premium to log-vs.-level outcome transformation is not reported and should be.

3. **Extend the sample to post-2017.** The 2010–2018 window misses the post-2017 Register of Contracts reform, a quasi-experimental shock to procurement transparency. Re-estimating the favoritism premium pre/post-2017 directly tests whether the transparency reform reduces the rent-extraction channel and would tighten the EU-cofunding heterogeneity story.

4. **Channel decomposition via auction red flags.** Decarolis and Giorgiantonio (2022) develop a battery of corruption indicators (single-bidding rate, runner-up gap, repeat-winner concentration) that distinguish bid-rigging from information-rent and from collusion. Running these on the Czech sample at low marginal cost would refine the 6% interpretation in advance of any structural model.

5. **Bid-rotation cartels.** The Bajari and Ye (2003) competition-vs.-collusion test, applied at the firm-pair level, can identify whether connected firms exhibit collusive bid-rotation patterns—in which case the 6% premium partially reflects a cartel rather than information rent.

6. **Heterogeneous EU-cofunding effect.** The Table 6 heterogeneity result deserves more analysis: is the moderator effect linear in oversight intensity, or threshold-shaped at a specific cofunding rate? A heterogeneous treatment-effect by EU-cofunding share would strengthen the policy-design claim by suggesting where to set the threshold.

7. **Long-run firm-growth outcomes.** Extending the close-bid winner-vs.-loser comparison to multi-year employment, revenue, and markup trajectories (Ferraz, Finan, and Szerman 2015) would surface the dynamic post-auction welfare consequences the static price-premium analysis cannot capture.

8. **Production-function sensitivity.** The paper maps a 6% price wedge directly to a 6% rent. Under imperfect competition this mapping is not exact (Pakes 2021): the wedge between price and marginal cost depends on the shape of the markup function. A brief sensitivity analysis with industry-level markup adjustments would tighten the welfare interpretation.

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